Rates of Change and Rectilinear Motion

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Position, Velocity & Speed**

A particle's velocity is given by $v(t) = 3t^2 - 12t + 9$ for $t \geq 0$. The position of the particle at time $t = 0$ is $s(0) = 5$. \\ Consider a second particle whose position $p(t)$ is described by the integral $p(t) = \int_0^t v(u)\,du$. \\ Find the net displacement of the first particle from $t = 1$ to $t = 3$.

Final Answer: -4. The net displacement of the first particle from $t = 1$ to $t = 3$ is given by the definite integral of its velocity function over this interval: $\Delta s = \int_1^3 v(t)\,dt$. \\ We are given $v(t) = 3t^2 - 12t + 9$. \\ To find the definite integral, we first find the antiderivative of $v(t)$: \\ $s(t) = \int (3t^2 - 12t + 9)\,dt = t^3 - 6t^2 + 9t + C$. \\ Since the initial position is $s(0) = 5$, we have $0^3 - 6(0)^2 + 9(0) + C = 5$, which means $C = 5$. Thus, $s(t) = t^3 - 6t^2 + 9t + 5$. \\ The net displacement is $s(3) - s(1)$. \\ $s(3) = (3)^3 - 6(3)^2 + 9(3) + 5 = 27 - 54 + 27 + 5 = 5$. \\ $s(1) = (1)^3 - 6(1)^2 + 9(1) + 5 = 1 - 6 + 9 + 5 = 9$. \\ Therefore, the net displacement is $s(3) - s(1) = 5 - 9 = -4$. \\ Alternatively, we can evaluate the definite integral directly: \\ $\int_1^3 (3t^2 - 12t + 9)\,dt = [t^3 - 6t^2 + 9t]_1^3$= [(3)^3 - 6(3)^2 + 9(3)] - [(1)^3 - 6(1)^2 + 9(1)] \\ $= [27 - 54 + 27] - [1 - 6 + 9]$= [0] - [4] = -4xp(t)) is extraneous and not needed to solve this problem.

2 **[Foundational]** • **Target: Total Distance vs. Displacement**

A particle's velocity is given by the function $v(t) = 5$ for $0 \leq t \leq 4$. Find the total displacement of the particle over this time interval.

Final Answer: 20. The displacement of the particle is the definite integral of its velocity function over the given time interval. \\ Displacement = $\int_0^4 v(t)\,dt = \int_0^4 5\,dt$. \\ To evaluate this definite integral, we find the antiderivative of 5, which is $5t$. \\ Then, we evaluate the antiderivative at the limits of integration: \\ $[5t]_0^4 = 5(4) - 5(0) = 20 - 0 = 20$. \\ Therefore, the total displacement of the particle is 20 units.

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3 [Foundational] • Target: Related Rates: Geometric Models

A particle's position in the first quadrant is described by a function $y(t)$, where t is time in seconds. The rate of change of its y-position with respect to time is governed by the differential equation $\frac{dy}{dt} = k\sin(t)$ for some positive constant k. The particle starts at the origin $(0, 0)$ at $t = 0$. At $t = \frac{\pi}{2}$ seconds, its y-position is $y(\frac{\pi}{2}) = 5$. \\ \\ Consider a second particle whose y-position $z(t)$ is described by the differential equation $\frac{dz}{dt} = k\cos(t)$, also starting at the origin at $t = 0$. If the first particle's y-position at $t = \frac{\pi}{4}$ is $y(\frac{\pi}{4}) = \frac{5}{\sqrt{2}}$, what is the value of k? Show your complete algebraic work.

Final Answer: 5. For the first particle, we have $\frac{dy}{dt} = k\sin(t)$. Integrating both sides with respect to t: \\ $y(t) = \int k\sin(t)\,dt = -k\cos(t) + C_1$ Since the particle starts at the origin, $y(0) = 0$. \\ x So, $y(t) = -k\cos(t) + k = k(1 - \cos(t))$. We are given that $y(\frac{\pi}{2}) = 5$. \\ x Therefore, the value of the constant k is 5. Let's verify with the information about the second particle. The second particle's y-position $z(t)$ is described by $\frac{dz}{dt} = k\cos(t)$. Integrating with respect to t: \\ $z(t) = \int k\cos(t)\,dt = k\sin(t) + C_2$ Since the second particle also starts at the origin, $z(0) = 0$. \\ x So, $z(t) = k\sin(t)$. Using the value of $k = 5$, we get $z(t) = 5\sin(t)$. Now, let's check the condition for the first particle at $t = \frac{\pi}{4}$: $y(\frac{\pi}{4}) = 5(1 - \cos(\frac{\pi}{4})) = 5(1 - \frac{\sqrt{2}}{2}) = 5 - \frac{5\sqrt{2}}{2}$. This does not match the given $y(\frac{\pi}{4}) = \frac{5}{\sqrt{2}} = \frac{5\sqrt{2}}{2}$. This indicates there might be a misunderstanding of the problem statement or a typo in the question's secondary condition. However, the value of k is uniquely determined by the first particle's trajectory information $y(0) = 0$ and $y(\frac{\pi}{2}) = 5$. The question asks for the value of k.

4 [Foundational] • Target: Related Rates: Angles & Proportions

A differentiable function $f(t)$ is represented by the graph below. What is the total change in $f(t)$ from $t = 2$ to $t = 5$? Show your complete algebraic work.

Final Answer: 1. The total change in a function $f(t)$ from $t = a$ to $t = b$ is given by the definite integral $\int_a^b f'(t)\,dt$. Alternatively, if the graph represents the function $f(t)$ itself, the total change is simply $f(b) - f(a)$. From the graph, we can observe that $f(2) = 1$ and $f(5) = 2$. Therefore, the total change is $f(5) - f(2) = 2 - 1 = 1$.

5 [Foundational] • Target: Position, Velocity & Speed

A particle moves with velocity $v(t) = 3t^2$. Displacement from $t = 0$ to $t = 2$ is:.

Applying Unit 8 · Applications of Integration principles, $\int_0^2 3t^2\,dt = [t^3]_0^2 = 8$ Therefore, the verified correct option is 8.

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6 [Foundational] • Target: Total Distance vs. Displacement

The graph of the derivative of a function $f(x)$, denoted by $f'(x)$, is shown in the figure above. The function $f(x)$ is continuous, and it is known that $f(1) = 5$. Determine the value of $f(5)$.

Final Answer: 14. The Fundamental Theorem of Calculus states that $\int_a^b f'(x)\,dx = f(b) - f(a)$. Rearranging this, we get $f(b) = f(a) + \int_a^b f'(x)\,dx$. In this problem, we are given $f(1) = 5$ and we need to find $f(5)$. Thus, we have $a = 1$ and $b = 5$. We need to calculate the definite integral $\int_1^5 f'(x)\,dx$, which represents the area under the curve of $f'(x)$ from $x = 1$ to $x = 5$. From the graph, $f'(x)$ is a straight line. The area from $x = 1$ to $x = 5$ can be split into two parts: a triangle and a rectangle. Part 1: Area from $x = 1$ to $x = 3$. This is a triangle with base $3 - 1 = 2$ and height $f'(1) = 3$. The area of this triangle is $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 2 \times 3 = 3$. Part 2: Area from $x = 3$ to $x = 5$. This is a rectangle with base $5 - 3 = 2$ and height $f'(3) = 3$. The area of this rectangle is $\text{base} \times \text{height} = 2 \times 3 = 6$. Therefore, the total area under the curve from $x = 1$ to $x = 5$ is $\int_1^5 f'(x)\,dx = 3 + 6 = 9$. Now, we can find $f(5)$: $f(5) = f(1) + \int_1^5 f'(x)\,dx = 5 + 9 = 14$. Common traps: Students might misinterpret the graph and assume $f'(x)$ is constant or linear over the entire interval without splitting it. They might also make calculation errors with the area of the triangle or rectangle.

7 [Foundational] • Target: Related Rates: Geometric Models

The graph of a function $f(x)$ is shown above. What is the instantaneous rate of change of $f(x)$ at $x = 2$? Show your complete algebraic work.

Final Answer: 0. The instantaneous rate of change of a function at a point is given by its derivative at that point, which corresponds to the slope of the tangent line to the graph of the function at that point. Observing the graph, the function $f(x)$ is a horizontal line segment at $y = 2$ for $0 \le x \le 4$. Therefore, the slope of this line segment is 0. Thus, the instantaneous rate of change of $f(x)$ at $x = 2$ is 0.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Related Rates: Angles & Proportions

Consider the function $f(x) = 4$. What is the value of the definite integral $\int_3^6 f(x)\,dx$? Show your complete algebraic work.

Final Answer: 12. The definite integral $\int_a^b c\,dx$ where c is a constant represents the area of a rectangle with height c and width $b - a$. In this case, $c = 4$, $a = 3$, and $b = 6$. Therefore, the value of the integral is $4 \times (6 - 3) = 4 \times 3 = 12$.

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9 **[Intermediate]** • **Target: Position, Velocity & Speed**

Area between $y = x^2$ and $y = x$ from 0 to 1 is:.

Final Answer: $\frac{1}{6}$. Applying Unit 6 · Integration and Accumulation of Change principles, $\int_0^1 (x - x^2)\,dx = \frac{1}{6}$ Therefore, the verified correct option is $\frac{1}{6}$.

10 **[Intermediate]** • **Target: Total Distance vs. Displacement**

Let R be the region bounded by the graph of $y = \sin(\pi x)$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the x-axis. The radius of a cross-section of the solid perpendicular to the x-axis is given by $r(x) = k\sin(\pi x)$ for some positive constant k. If the volume of the solid is $\frac{\pi}{2}$ cubic units, determine the value of k.

Final Answer: 1. The volume of a solid of revolution generated by revolving a region bounded by $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$ about the x-axis is given by the disk method formula $V = \int_a^b \pi[f(x)]^2\,dx$. In this problem, the region R is bounded by $y = \sin(\pi x)$, the x-axis, $x = 0$, and $x = 1$. The radius of a cross-section is given by $r(x) = k\sin(\pi x)$. Therefore, the volume of the solid is $V = \int_0^1 \pi[r(x)]^2\,dx = \int_0^1 \pi(k\sin(\pi x))^2\,dx$. We are given that the volume is $\frac{\pi}{5}$. So, we have the equation: $\int_0^1 \pi k^2 \sin^2(\pi x)\,dx = \frac{\pi}{2}$ We can cancel π from both sides: $k^2 \int_0^1 \sin^2(\pi x)\,dx = \frac{1}{2}$ To evaluate the integral $\int_0^1 \sin^2(\pi x)\,dx$, we use the power-reduction identity $\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$. Let $\theta = \pi x$, so $2\theta = 2\pi x$. Thus, $\sin^2(\pi x) = \frac{1 - \cos(2\pi x)}{2}$. $\int_0^1 \frac{1 - \cos(2\pi x)}{2}\,dx = \frac{1}{2}\int_0^1 (1 - \cos(2\pi x))\,dx$ Now, we integrate: $\frac{1}{2}\left[x - \frac{\sin(2\pi x)}{2\pi}\right]_0^1$ Evaluate at the limits: $\frac{1}{2}\left[\left(1 - \frac{\sin(2\pi)}{2\pi}\right) - \left(0 - \frac{\sin(0)}{2\pi}\right)\right]$ Since $\sin(2\pi) = 0$ and $\sin(0) = 0$, this simplifies to: $\frac{1}{2}[(1 - 0) - (0 - 0)] = \frac{1}{2}$ So, the integral evaluates to $\frac{1}{2}$. Substitute this back into our equation for k^2: $k^2\left(\frac{1}{2}\right) = \frac{1}{2}$ Solving for k^2: $k^2 = 1$ Since k is a positive constant, we take the positive square root: $k = 1$

11 **[Intermediate]** • **Target: Related Rates: Geometric Models**

The graph of the function $f(x) = ax - 3$ is shown below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 0$ to $x = 4$ is 20 square units. Find the value of the parameter a.

Final Answer: 4. The function is given by $f(x) = ax - 3$. The area of the shaded region is represented by the definite integral of $f(x)$ from $x = 0$ to $x = 4$. We are given that this area is 20. $\int_0^4 (ax - 3)\,dx = 20$ First, find the antiderivative of $ax - 3$: $\int (ax - 3)\,dx = \frac{a}{2}x^2 - 3x + C$ Now, evaluate the definite integral from 0 to 4: $\left[\frac{a}{2}x^2 - 3x\right]_0^4 = \left(\frac{a}{2}(4)^2 - 3(4)\right) - \left(\frac{a}{2}(0)^2 - 3(0)\right) = \left(\frac{a}{2}(16) - 12\right) - (0) = 8a - 12$ We are given that this area is 20: $8a - 12 = 20$ Now, solve for a: $8a = 20 + 12$ $8a = 32$ $a = \frac{32}{8}$ $a = 4$ Therefore, the value of the parameter a is 4.

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12 **[Intermediate]** • **Target: Related Rates: Angles & Proportions**

The velocity of a particle moving along a straight line is given by the function $v(t) = t^2 - 6t + 8$ m/s. What is the instantaneous acceleration of the particle at time $t = 3$ seconds? Show your complete algebraic work.

Final Answer: -2 m/s^2. The velocity of the particle is given by $v(t) = t^2 - 6t + 8$. Acceleration is the rate of change of velocity with respect to time, which means acceleration is the derivative of the velocity function, $a(t) = v'(t)$. Differentiating $v(t)$ with respect to t, we get: $a(t) = \frac{d}{dt}(t^2 - 6t + 8) = 2t - 6$. We need to find the instantaneous acceleration at time $t = 3$ seconds. Substituting $t = 3$ into the acceleration function: $a(3) = 2(3) - 6 = 6 - 6 = 0$ m/s^2. Therefore, the instantaneous acceleration of the particle at time $t = 3$ seconds is 0 m/s^2.

13 **[Intermediate]** • **Target: Position, Velocity & Speed**

A particle moves along a straight line so that its velocity at time $t \geq 0$ is modeled by $v(t) = 6t^2 - 18t + 12$. At what time t is the acceleration of the particle equal to 0? Show your complete algebraic work.

Final Answer: $t = \frac{3}{2}$. To find the acceleration $a(t)$, take the derivative of the velocity function $v(t)$ with respect to time t: $a(t) = v'(t) = \frac{d}{dt}(6t^2 - 18t + 12) = 12t - 18$. Set $a(t) = 0$ and solve for t: $12t - 18 = 0 \Rightarrow 12t = 18 \Rightarrow t = \frac{18}{12} = \frac{3}{2}$. Common Misconceptions: - Solving $v(t) = 0$ yields $6(t - 1)(t - 2) = 0 \Rightarrow t = 1$ or $t = 2$. These are the times when the particle is at rest, not when its acceleration is zero. - Omitting the factor of 2 when applying the power rule gives $6t - 18 = 0 \Rightarrow t = 3$.

14 **[Intermediate]** • **Target: Total Distance vs. Displacement**

A differentiable function $f(x)$ is given by the equation $f(x) = x^2 - 6x + 10$. Find the instantaneous rate of change of $f(x)$ at $x = 4$.

Final Answer: 2. The instantaneous rate of change of a function at a point is given by its derivative evaluated at that point. First, we find the derivative of $f(x) = x^2 - 6x + 10$. Using the power rule, $f'(x) = 2x - 6$. Next, we evaluate the derivative at $x = 4$: $f'(4) = 2(4) - 6 = 8 - 6 = 2$. Therefore, the instantaneous rate of change of $f(x)$ at $x = 4$ is 2.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Related Rates: Geometric Models • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Consider the function $f(x) = 3x^2 - 6x + 5$. What is the instantaneous rate of change of $f(x)$ at $x = 2$? Show your complete algebraic work.

Final Answer: 6. The instantaneous rate of change of a function at a point is given by its derivative evaluated at that point. First, find the derivative of $f(x)$: $f'(x) = \frac{d}{dx}(3x^2 - 6x + 5) = 6x - 6$. Next, evaluate $f'(x)$ at $x = 2$: $f'(2) = 6(2) - 6 = 12 - 6 = 6$. Therefore, the instantaneous rate of change of $f(x)$ at $x = 2$ is 6.

16 [Challenging] • Target: Related Rates: Angles & Proportions • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

$\int e^x \, dx = .$

Final Answer: $e^x + C$. Applying Unit 6 · Integration and Accumulation of Change principles, e^x is its own antiderivative. Therefore, the verified correct option is $e^x + C$.

17 [Challenging] • Target: Position, Velocity & Speed • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

$\int_1^4 \frac{1}{x} \, dx = .$

Final Answer: $\ln 4$. Applying Unit 6 · Integration and Accumulation of Change principles, $[\ln x]_1^4 = \ln 4 - \ln 1 = \ln 4$ Therefore, the verified correct option is $\ln 4$.

18 [Challenging] • Target: Total Distance vs. Displacement • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

$\int 4x^3 \, dx = .$

Final Answer: $x^4 + C$. Applying Unit 6 · Integration and Accumulation of Change principles, $\frac{4x^4}{4} + C = x^4 + C$ Therefore, the verified correct option is $x^4 + C$.

19 [Challenging] • Target: Related Rates: Geometric Models • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

$\int_0^\pi \sin x \, dx = .$

Final Answer: 2. Applying Unit 6 · Integration and Accumulation of Change principles, $[-\cos x]_0^\pi = 1 + 1 = 2$ Therefore, the verified correct option is 2.

20 [Challenging] • Target: Related Rates: Angles & Proportions • Cross-Lesson: 7.01 Differential Equations and Slope Fields

$\int_0^3 2x \, dx = .$

Final Answer: 9. Applying Unit 6 · Integration and Accumulation of Change principles, $[x^2]_0^3 = 9 - 0 = 9$ Therefore, the verified correct option is 9.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Position, Velocity & Speed

A particle moves along the x-axis with position $x(t) = t^3 - 6t^2 + 9t$ for $t \geq 0$. At what times is the particle at rest?

a. $t = 1$ **and** $t = 3$ ✓

b. $t = 0$ and $t = 3$

c. $t = 2$

d. $t = 3$ only

At rest means velocity $v(t) = x'(t) = 0$: $3t^2 - 12t + 9 = 0 \Rightarrow 3(t-1)(t-3) = 0 \Rightarrow t = 1, 3$.

22 [Foundational] • Target: Position, Velocity & Speed

What is the speed of the particle when $v(t) = -8$ and $a(t) = 3$?

a. 8 ✓

b. -8

c. 5

d. $\sqrt{73}$

Speed is defined as the absolute value of velocity: $|v(t)| = |-8| = 8$.

23 [Foundational] • Target: Position, Velocity & Speed

When is a particle's speed increasing?

a. **When velocity and acceleration have the same sign** ✓

b. When velocity is positive

c. When acceleration is positive

d. When acceleration is zero

Speed $|v(t)|$ increases when velocity and acceleration work in the same direction, meaning $v(t)$ and $a(t)$ have the same sign.

24 [Foundational] • Target: Total Distance vs. Displacement

If a particle's position is $s(t) = t^2 - 4t$ on $[0, 3]$, what is its net displacement?

a. -3 ✓

b. 5

c. -4

d. 3

Displacement $= s(3) - s(0) = (9 - 12) - 0 = -3$.

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25 [Foundational] • Target: Total Distance vs. Displacement

For the above particle with $s(t) = t^2 - 4t$ on $[0, 3]$, what is the total distance traveled?

a. 5 ✓

b. 3

c. 4

d. 7

$v(t) = 2t - 4 = 0 \Rightarrow t = 2$. From $t = 0$ to 2: $|s(2) - s(0)| = |-4 - 0| = 4$. From $t = 2$ to 3: $|s(3) - s(2)| = |-3 - (-4)| = 1$. Total distance $= 4 + 1 = 5$.

26 [Foundational] • Target: Total Distance vs. Displacement

Which integral represents the total distance traveled by a particle from $t = a$ to $t = b$?

a. $\int_a^b |v(t)|dt$ ✓

b. $\int_a^b v(t)dt$

c. $|\int_a^b v(t)dt|$

d. $\int_a^b a(t)dt$

Total distance accumulates the speed over time: $\int_a^b |v(t)|dt$.

27 [Foundational] • Target: Related Rates: Geometric Models

The radius of a circle increases at a constant rate of 3 cm/s. How fast is the area increasing when the radius is 10 cm?

a. 60π **cm²/s** ✓

b. 30π cm²/s

c. 100π cm²/s

d. 20π cm²/s

$A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi(10)(3) = 60\pi$ cm²/s.

28 [Foundational] • Target: Related Rates: Geometric Models

A 13-foot ladder leans against a vertical wall. If the bottom slides away from the wall at 2 ft/s, how fast is the top sliding down when the bottom is 5 feet from the wall?

a. $-\frac{5}{6}$ **ft/s** ✓

b. $-\frac{12}{5}$ ft/s

c. -2 ft/s

d. $-\frac{5}{12}$ ft/s

$x^2 + y^2 = 13^2 = 169$. When $x = 5$, $y = 12$. Differentiating: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow 5(2) + 12\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{10}{12} = -\frac{5}{6}$ ft/s.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Related Rates: Geometric Models**

Air is pumped into a spherical balloon at a rate of 100 cm^3/s. How fast is the radius increasing when $r = 5$ cm?

a. $\frac{1}{\pi}$ **cm/s ✓**

b. $\frac{1}{4\pi}$ cm/s

c. $\frac{4}{\pi}$ cm/s

d. 25π cm/s

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} \Rightarrow 100 = 4\pi(25)\frac{dr}{dt} \Rightarrow 100 = 100\pi\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{\pi}$ cm/s.

30 **[Intermediate]** • **Target: Related Rates: Angles & Proportions**

A conical tank with vertex down has height 10 m and base radius 4 m. Water flows in at 2 m^3/min. What is the relation between r and h?

a. $r = \frac{2}{5}h$ **✓**

b. $r = \frac{5}{2}h$

c. $r = \frac{1}{2}h$

d. $r = 4h$

By similar triangles in the cross-section: $\frac{r}{h} = \frac{4}{10} = \frac{2}{5} \Rightarrow r = \frac{2}{5}h$.

31 **[Intermediate]** • **Target: Related Rates: Angles & Proportions**

Using $V = \frac{1}{3}\pi r^2 h = \frac{4}{75}\pi h^3$, how fast is the water level rising when $h = 5$ m and $\frac{dV}{dt} = 2$ m^3/min?

a. $\frac{1}{2\pi}$ **m/min ✓**

b. $\frac{2}{\pi}$ m/min

c. $\frac{1}{4\pi}$ m/min

d. $\frac{4}{25\pi}$ m/min

$\frac{dV}{dt} = \frac{4}{25}\pi h^2 \frac{dh}{dt} \Rightarrow 2 = \frac{4}{25}\pi(25)\frac{dh}{dt} = 4\pi\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{2}{4\pi} = \frac{1}{2\pi}$ m/min.

32 **[Intermediate]** • **Target: Related Rates: Angles & Proportions**

A street light is mounted at the top of a 15-foot pole. A 6-foot man walks away from the pole at 5 ft/s. How fast is the tip of his shadow moving?

a. $\frac{25}{3}$ **ft/s ✓**

b. 5 ft/s

c. $\frac{10}{3}$ ft/s

d. 10 ft/s

Let s be shadow tip distance: $\frac{s}{15} = \frac{s-x}{6} \Rightarrow 6s = 15s - 15x \Rightarrow 9s = 15x \Rightarrow s = \frac{5}{3}x$. Then $\frac{ds}{dt} = \frac{5}{3}\frac{dx}{dt} = \frac{5}{3}(5) = \frac{25}{3}$ ft/s.

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33 [Intermediate] • Target: Position, Velocity & Speed

A particle moves along the x-axis. Its velocity is given by the function $v(t) = \sqrt{t^2 + 1}$ for $t \geq 0$. The position of the particle at time $t = 0$ is $x(0) = 5$. Consider the function $P(x)$ defined as the total displacement of the particle from $t = 0$ to $t = x$. That is, $P(x) = x(x) - x(0)$. Which of the following statements about $P(x)$ is true for $x > 0$?

a. $P(x)$ **is always positive and increasing.** ✓

b. $P(x)$ is always negative and decreasing.

c. $P(x)$ is always positive and decreasing.

d. $P(x)$ is always negative and increasing.

The velocity of the particle is given by $v(t) = \sqrt{t^2 + 1}$. Since $t \geq 0$, $t^2 \geq 0$, so $t^2 + 1 \geq 1$. Therefore, $v(t) = \sqrt{t^2 + 1} \geq \sqrt{1} = 1$ for all $t \geq 0$. This means the velocity is always positive. The position of the particle is given by $x(t)$. The Fundamental Theorem of Calculus states that the change in position (displacement) is the integral of the velocity function: $x(t) - x(0) = \int_0^t v(u)\,du$. The function $P(x)$ represents the total displacement from $t = 0$ to $t = x$: $P(x) = x(x) - x(0) = \int_0^x \sqrt{t^2 + 1}\,dt$. Since $v(t) = \sqrt{t^2 + 1}$ is always positive for $t \geq 0$, its definite integral from 0 to x (where $x > 0$) must be positive. Thus, $P(x) > 0$ for $x > 0$. To determine if $P(x)$ is increasing or decreasing, we examine its derivative with respect to x. By the Fundamental Theorem of Calculus, $P'(x) = \frac{d}{dx}\left(\int_0^x \sqrt{t^2 + 1}\,dt\right) = \sqrt{x^2 + 1}$. Since $x > 0$, $x^2 > 0$, so $x^2 + 1 > 1$, which means $P'(x) = \sqrt{x^2 + 1} > 1$. A positive derivative indicates that the function is increasing. Therefore, $P(x)$ is always positive and increasing for $x > 0$.

34 [Intermediate] • Target: Total Distance vs. Displacement

Given $\int_{-2}^{-1} 1x^2\,dx = \frac{7}{3}$ and $\int_{-1}^{0} 1x^2\,dx = \frac{1}{3}$, find $\int_{-2}^{0} 1x^2\,dx$.

a. $-\frac{8}{3}$

b. $\frac{1}{3}$

c. $\frac{8}{3}$ ✓

d. $\frac{7}{3}$

By additivity of the definite integral: $\int_{-2}^{0} = \int_{-2}^{-1} + \int_{-1}^{0} = \frac{7}{3} + \frac{1}{3} = \frac{8}{3}$.

35 [Intermediate] • Target: Related Rates: Geometric Models

Given $\int_{-1}^{1} 2x^2\,dx = \frac{4}{3}$ and $\int_{1}^{3} 2x^2\,dx = \frac{52}{3}$, find $\int_{-1}^{3} 2x^2\,dx$.

a. $\frac{4}{3}$

b. $\frac{52}{3}$

c. $-\frac{56}{3}$

d. $\frac{56}{3}$ ✓

By additivity of the definite integral: $\int_{-1}^{3} = \int_{-1}^{1} + \int_{1}^{3} = \frac{4}{3} + \frac{52}{3} = \frac{56}{3}$.

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36 [Intermediate] • Target: Related Rates: Angles & Proportions

Given $\int_0^1 2x^2\,dx = \frac{2}{3}$ and $\int_1^2 2x^2\,dx = \frac{14}{3}$, find $\int_0^2 2x^2\,dx$.

a. $\frac{14}{3}$

b. $\frac{2}{3}$

c. $\frac{16}{3}$ ✓

d. $-\frac{16}{3}$

By additivity of the definite integral: $\int_0^2 = \int_0^1 + \int_1^2 = \frac{2}{3} + \frac{14}{3} = \frac{16}{3}$.

37 [Intermediate] • Target: Position, Velocity & Speed

Given $\int_{-1}^0 4x^2\,dx = \frac{4}{3}$ and $\int_0^1 4x^2\,dx = \frac{4}{3}$, find $\int_{-1}^1 4x^2\,dx$.

a. $\frac{4}{3}$

b. $\frac{8}{3}$ ✓

c. $-\frac{8}{3}$

d. $\frac{17}{3}$

By additivity of the definite integral: $\int_{-1}^1 = \int_{-1}^0 + \int_0^1 = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Total Distance vs. Displacement • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Given $\int_{-2}^1 5x^2\,dx = 15$ and $\int_1^5 5x^2\,dx = \frac{620}{3}$, find $\int_{-2}^5 5x^2\,dx$.

a. $\frac{620}{3}$

b. 15

c. $-\frac{665}{3}$

d. $\frac{665}{3}$ ✓

By additivity of the definite integral: $\int_{-2}^5 = \int_{-2}^1 + \int_1^5 = 15 + \frac{620}{3} = \frac{665}{3}$.

39 [Challenging] • Target: Related Rates: Geometric Models • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Given $\int_{-2}^1 4x^2\,dx = 12$ and $\int_1^3 4x^2\,dx = \frac{104}{3}$, find $\int_{-2}^3 4x^2\,dx$.

a. 12

b. $\frac{104}{3}$

c. $-\frac{140}{3}$

d. $\frac{140}{3}$ ✓

By additivity of the definite integral: $\int_{-2}^3 = \int_{-2}^1 + \int_1^3 = 12 + \frac{104}{3} = \frac{140}{3}$.

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40 [Challenging] • Target: Related Rates: Angles & Proportions • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Given $\int_{-1}^{2} 4x^2\,dx = 12$ and $\int_{2}^{6} 4x^2\,dx = \frac{832}{3}$, find $\int_{-1}^{6} 4x^2\,dx$.

a. 12

b. $\frac{868}{3}$ ✓

c. $-\frac{868}{3}$

d. $\frac{832}{3}$

By additivity of the definite integral: $\int_{-1}^{6} = \int_{-1}^{2} + \int_{2}^{6} = 12 + \frac{832}{3} = \frac{868}{3}$.

41 [Challenging] • Target: Position, Velocity & Speed • Cross-Lesson: 8.01 Area Between Curves

Given $\int_{-3}^{-1} 5x^2\,dx = \frac{130}{3}$ and $\int_{-1}^{0} 5x^2\,dx = \frac{5}{3}$, find $\int_{-3}^{0} 5x^2\,dx$.

a. $\frac{130}{3}$

b. -45

c. **45** ✓

d. $\frac{5}{3}$

By additivity of the definite integral: $\int_{-3}^{0} = \int_{-3}^{-1} + \int_{-1}^{0} = \frac{130}{3} + \frac{5}{3} = 45$.

42 [Challenging] • Target: Total Distance vs. Displacement • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Given $\int_{-2}^{0} 5x^2\,dx = \frac{40}{3}$ and $\int_{0}^{4} 5x^2\,dx = \frac{320}{3}$, find $\int_{-2}^{4} 5x^2\,dx$.

a. **120** ✓

b. $\frac{320}{3}$

c. -120

d. $\frac{40}{3}$

By additivity of the definite integral: $\int_{-2}^{4} = \int_{-2}^{0} + \int_{0}^{4} = \frac{40}{3} + \frac{320}{3} = 120$.

43 [Challenging] • Target: Related Rates: Geometric Models • Cross-Lesson: 1.01 Limits Graphically and Numerically

Given $\int_{-3}^{-2} 2x^2\,dx = \frac{38}{3}$ and $\int_{-2}^{0} 2x^2\,dx = \frac{16}{3}$, find $\int_{-3}^{0} 2x^2\,dx$.

a. $\frac{38}{3}$

b. -18

c. **18** ✓

d. $\frac{16}{3}$

By additivity of the definite integral: $\int_{-3}^{0} = \int_{-3}^{-2} + \int_{-2}^{0} = \frac{38}{3} + \frac{16}{3} = 18$.

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44 [Challenging] • Target: Related Rates: Angles & Proportions • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Given $\int_{-1}^{2} 2x^2\,dx = 6$ and $\int_{2}^{6} 2x^2\,dx = \frac{416}{3}$, find $\int_{-1}^{6} 2x^2\,dx$.

a. $-\frac{434}{3}$

b. 6

c. $\frac{416}{3}$

d. $\frac{434}{3}$ ✓

By additivity of the definite integral: $\int_{-1}^{6} = \int_{-1}^{2} + \int_{2}^{6} = 6 + \frac{416}{3} = \frac{434}{3}$.

45 [Challenging] • Target: Position, Velocity & Speed • Cross-Lesson: 2.01 The Definition of the Derivative

Given $\int_{-2}^{0} 4x^2\,dx = \frac{32}{3}$ and $\int_{0}^{3} 4x^2\,dx = 36$, find $\int_{-2}^{3} 4x^2\,dx$.

a. $\frac{140}{3}$ ✓

b. 36

c. $\frac{32}{3}$

d. $-\frac{140}{3}$

By additivity of the definite integral: $\int_{-2}^{3} = \int_{-2}^{0} + \int_{0}^{3} = \frac{32}{3} + 36 = \frac{140}{3}$.

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